Moeez Ishaq

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Education

Wilfrid Laurier University — Waterloo, ON
B.Sc. Computer Science & BBA (Double Degree) | GPA: 3.8/4.0 | Dean's List — *September 2023 – April 2028*

Experience

Machine Learning Engineer Co-op (AI/Software) — September 2026 – Present
Royal Bank of Canada (RBC) — *Toronto, ON*

- Reduced manual AI-assisted development tracking by building a **Spec Kit extension** with **16 lifecycle hooks** that auto-captures **token, cost, and duration** data across **8 SDLC phases**, syncing to **GA4**
- Improved analytics reliability by implementing orphaned-phase reconciliation, duplicate-run detection, and a 4-hour auto-abandon safeguard in a **Bash/Python** pipeline, validated by a **61-test regression suite**
- Gave teams visibility into AI versus human development costs trade-offs with a self-contained **HTML dashboard** of per-phase, per-model, and per-story breakdowns across **16 LLM pricing models**
- Contributed to an internal adoption portal (**FastAPI, React/TypeScript, MongoDB**) that scores **15,000+ repositories** across 6 adoption tiers, rolling results up 4 org levels, giving leadership one view of SpecKit adoption

Software Engineer Co-op — January 2026 – April 2026
Royal Bank of Canada (RBC) — *Toronto, ON*

- Automated documentation for **47 internal APIs** with a **JavaScript pipeline** that generates LLM-based docs and publishes to **Confluence via MCP**, supporting an AI initiative **estimated to save ∼$700K annually**
- Reduced vulnerability remediation to a **single PR merge** by building a **Python pipeline** using **LangChain** and an **internal LLM gateway** that automatically detects, fixes, tests, and raises pull requests for security alerts
- Eliminated **100% of developer effort** triggering batch jobs with a self-service platform for **Excel/CSV → JSON** transformations across **1000+ record batches** of KYC and financial data, backed by **MongoDB** and **AWS S3**
- Reduced **Client ID Visualizer** response time by **∼50%** by parallelizing **GraphQL** and **REST** calls with **CompletableFuture**, deployed on **OpenShift/Kubernetes** powering **1000+ searches**

Data Engineer Co-op — May 2025 – August 2025
Mattamy Homes — *Mississauga, ON*

- Resolved Spark driver OOM crashes by rewriting **R reshape2** melt in **PySpark**, saving **30–40 minutes per run**
- Engineered **Python** ingestion pipelines in **Databricks** to collect and normalize housing data from multiple real estate boards, feeding **45,000+ records monthly** into a data lake that powered national reporting dashboards
- Automated weekly ingestion of **50,000+ housing records** via **Azure Data Factory and SQL ETL pipelines**

Projects

AI-Powered Navigation Assistant, *Carnegie Mellon University NexHacks 2026 | TypeScript, React, LLMs*

- Enabled **voice-guided indoor navigation for visually impaired users** by converting speech-to-text queries into adaptive, step-by-step routes via **Gemini LLM** and **ElevenLabs** text-to-speech
- Warned visually impaired users of obstacles **within 2 seconds** of detection by combining **Overshoot AI** with **LLM reasoning** over structured floor plan data to trigger live voice alerts

Neural Network-Based Music Sequence Predictor | *Python, TensorFlow, Keras, NumPy, Pandas*

- Trained a **dilated Conv1D + LSTM** model in **TensorFlow/Keras** on 4-voice Bach chorales (NumPy/Pandas pipeline), reaching **∼82% validation next-note accuracy** and generating new chorales from an 8-chord seed

Stocks Fantasy League | *TypeScript, JavaScript, Node.js, React, Supabase, PostgreSQL, Firebase*

- Drove **stock portfolio valuations and league rankings** by building a **TypeScript/React** platform integrating **Alpha Vantage API** hourly market data, backed by **Supabase/PostgreSQL** and **Firebase** authentication

Technical Skills

Languages: Python, JavaScript, Java, SQL, R, TypeScript, GraphQL
Frameworks & Technologies: React, Node.js, Spring Boot, FastAPI, PySpark, TensorFlow/Keras, NumPy/Pandas, AWS S3, Microsoft Azure, Databricks, OpenShift/Kubernetes, CI/CD, PostgreSQL, MySQL, Supabase, MongoDB
AI Tools: Claude Code, GitHub Copilot, Cursor, Devin (CLI, Desktop, Cloud), LLM APIs, MCP, LangChain
Certifications: JPMorgan Chase Software Engineer Forage Simulation, Databricks Certified, Salesforce Certified x4